

Module 9

[Advanced Topics: Java Swing Programming]

Topic 1[java swing: Money Changer]

Problem Statement: Write a Money Changer (Dollar to Taka) program in Java with the following components. When user clicks **Convert** button, its shows the output in a label.

Input \$

3

Tk 310

Convert

Code:

```
public class MoneyChanger {
    public static void main(String[] args) {
        javax.swing.JFrame frame = new javax.swing.JFrame("Money Changer");
        frame.setSize(350, 220);
        frame.setDefaultCloseOperation(javax.swing.JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);

        javax.swing.JLabel inputLabel = new javax.swing.JLabel("Input $");
        inputLabel.setBounds(60, 40, 80, 25);
        frame.add(inputLabel);

        javax.swing.JTextField dollarField = new javax.swing.JTextField();
        dollarField.setBounds(60, 70, 80, 30);
        frame.add(dollarField);

        javax.swing.JLabel resultLabel = new javax.swing.JLabel("Tk 0");
        resultLabel.setBounds(200, 75, 100, 25);
        frame.add(resultLabel);

        javax.swing.JButton convertButton = new javax.swing.JButton("Convert");
        convertButton.setBounds(120, 130, 100, 35);
        frame.add(convertButton);

        convertButton.addActionListener(e -> {
            try {
                double dollar = Double.parseDouble(dollarField.getText());
                double taka = dollar * 110;    // 1 Dollar = 110 Taka
                resultLabel.setText("Tk " + (int)taka);
            } catch (NumberFormatException ex) {
```

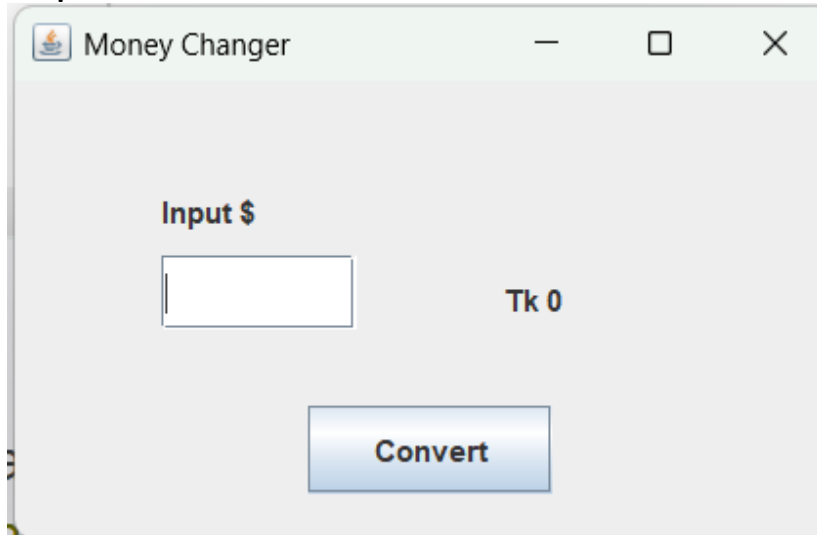
```

        resultLabel.setText("Invalid Input");
    }
});

frame.setLocationRelativeTo(null);
frame.setVisible(true);
}
}

```

Output:



Topic 2[java swing: Login Form]

Problem Statement: Write a Java program that accepts username and password of user and then displays whether the user is valid or NOT., It shows the output in a label.

Login Form	
Username	
Password	
Submit	
Valid User	

Code:

```

public class LoginForm {
    public static void main(String[] args) {
        javax.swing.JFrame frame = new javax.swing.JFrame("Login Form");
        frame.setSize(360, 250);
        frame.setDefaultCloseOperation(javax.swing.JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);

        javax.swing.JLabel titleLabel = new javax.swing.JLabel("Login Form");
        titleLabel.setBounds(30, 10, 100, 25);
    }
}

```

```

frame.add(titleLabel);

javax.swing.JLabel userLabel = new javax.swing.JLabel("Username");
userLabel.setBounds(40, 60, 80, 25);
frame.add(userLabel);

javax.swing.JTextField userField = new javax.swing.JTextField();
userField.setBounds(130, 60, 140, 25);
frame.add(userField);

javax.swing.JLabel passLabel = new javax.swing.JLabel("Password");
passLabel.setBounds(40, 100, 80, 25);
frame.add(passLabel);

javax.swing.JPasswordField passField = new javax.swing.JPasswordField();
passField.setBounds(130, 100, 140, 25);
frame.add(passField);

javax.swing.JButton submitButton = new javax.swing.JButton("Submit");
submitButton.setBounds(120, 145, 90, 35);
frame.add(submitButton);

javax.swing.JLabel resultLabel = new javax.swing.JLabel("");
resultLabel.setBounds(120, 185, 120, 25);
frame.add(resultLabel);

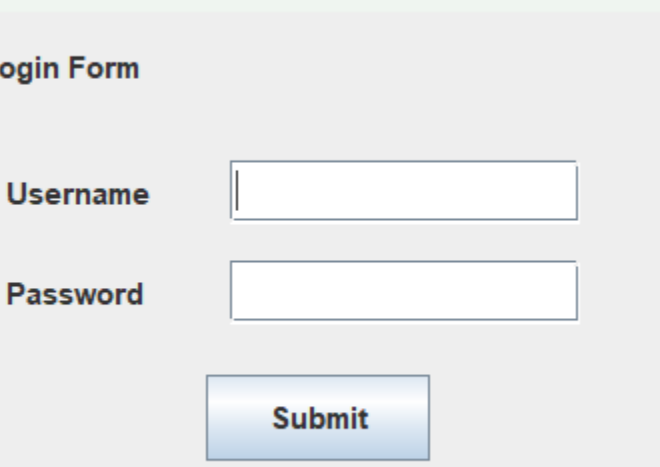
submitButton.addActionListener(e -> {
    String username = userField.getText();
    String password = new String(passField.getPassword());

    if (username.equals("admin") && password.equals("1234")) {
        resultLabel.setText("Valid User");
    } else {
        resultLabel.setText("Invalid User");
    }
});

frame.setLocationRelativeTo(null);
frame.setVisible(true);
}
}

```

Output:



Login Form

Username

Password

Submit